miniature disks or Frisbees, or by three-dimensional bloblike ingredients, shaped like baseballs or lumps of clay. Or, since the theory has such an abundance of space dimensions, we can even imagine blobs with more dimensions still. Why don't these ingredients play any role in our fundamental theories?

In the 1980s and early 1990s, most string theorists had what seemed like a convincing answer. They argued that there *had* been attempts to formulate a fundamental theory of matter based on bloblike constituents by, among others, such icons of twentieth-century physics as Werner Heisenberg and Paul Dirac. But their work, as well as many subsequent studies, showed that it was extremely difficult to develop a theory based on tiny blobs that met the most basic of physical requirements—for example, ensuring that all quantum mechanical probabilities lie between 0 and 1 (no sense can be made of negative probabilities or of probabilities greater than 1), and debarring faster-than-light communication. For point particles, a half-century of research initiated in the 1920s showed that these conditions could be met (as long as gravity was ignored). And, by the 1980s, more than a decade of investigation by Schwarz, Scherk, Green, and others established, to the surprise of most researchers, that the conditions could also be met for one-dimensional ingredients, strings (which necessarily *included* gravity). But it seemed impossible to proceed to fundamental ingredients with two or more spatial dimensions. The reason, briefly put, is that the number of symmetries respected by the equations peaks enormously for one-dimensional objects (strings) and drops off precipitously thereafter. The symmetries in question are more abstract than the ones discussed in Chapter 8 (they have to do with how equations change if, while studying the motion of a string or a higher dimensional ingredient, we were to zoom in or out, suddenly and arbitrarily changing the resolution of our observations). These transformations prove critical to formulating a physically sensible set of equations, and beyond strings it seemed that the required fecundity of symmetries was absent.[1]

It was thus another shock to most string theorists when Witten's paper and an avalanche of subsequent results[2] led to the realization that string theory, and the M-theoretic framework to which it now belongs, *does* contain ingredients besides strings. The analyses showed that there are two-dimensional objects called, naturally enough, *membranes* (another